Green University of Bangladesh
Department of Computer Science and Engineering(CSE)

Faculty of Sciences and Engineering

Semester: (Spring , Year:2025), B.Sc. in CSE (Day)

LAB REPORT NO 4

Course Title:Database Lab

Course Code:CSE 210 Section:242-D3

Lab Experiment Name: Implementation of Triggers for Salary Automation and Logging

Student Details

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[For Teachers use only: Don't Write Anything inside this box]

Lab Report Status

Marks: …… Signature:.....

Comments:..... Date:.....

1. Objectives

- To implement **BEFORE UPDATE** triggers for business logic automation.
- To implement **AFTER UPDATE** triggers for auditing and logging purposes.
- To manage data integrity using constraints and conditional logic within the database.

2. Database Schema

The implementation consists of two tables:

- **EMPLOYEE:** Stores employee personal information, publication counts, and salary details.
- **SALARY_LOG:** Stores the historical trail of salary adjustments.

3. SQL Implementation

Database and Table Initialization

Server: 127.0.0.1

Databases | SQL | Status | User accounts | Export | Imp

Run SQL query/queries on server "127.0.0.1":

```sql
DROP DATABASE IF EXISTS faculty_trigger_lab;
CREATE DATABASE faculty_trigger_lab;
USE faculty_trigger_lab;
CREATE TABLE EMPLOYEE (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(100) NOT NULL,
    BasicSalary DECIMAL(10,2) NOT NULL,
    StartDate DATE NOT NULL,
    NoOfPub INT NOT NULL CHECK(NoOfPub >= 0),
    IncrementRate DECIMAL(5,2) DEFAULT 0,
    UpdatedSalary DECIMAL(10,2)
);

CREATE TABLE SALARY_LOG (
    LogID INT AUTO_INCREMENT PRIMARY KEY,
    EmpID INT,
    OldSalary DECIMAL(10,2),
    NewSalary DECIMAL(10,2),
    ChangedAt DATETIME DEFAULT CURRENT_TIMESTAMP,
    Note VARCHAR(100),
    FOREIGN KEY (EmpID) REFERENCES EMPLOYEE(EmpID)
);
```

Logic Trigger (BEFORE UPDATE)

Server: 127.0.0.1

Databases | SQL | Status | User accounts | Export | Import | Settings | R

```sql
CREATE TRIGGER trg_calculate_salary
BEFORE UPDATE ON EMPLOYEE
FOR EACH ROW
BEGIN
    DECLARE duration INT;
    SET duration = TIMESTAMPDIFF(YEAR, OLD.StartDate, CURDATE());

    IF duration > 1 THEN
        IF NEW.NoOfPub > 4 THEN
            SET NEW.IncrementRate = 20.00;
        ELSEIF NEW.NoOfPub IN (2, 3) THEN
            SET NEW.IncrementRate = 10.00;
        ELSEIF NEW.NoOfPub = 1 THEN
            SET NEW.IncrementRate = 5.00;
        ELSE
            SET NEW.IncrementRate = 0.00;
        END IF;
    ELSE
        SET NEW.IncrementRate = 0.00;
    END IF;

    SET NEW.UpdatedSalary = OLD.BasicSalary + (OLD.BasicSalary * (NEW.IncrementRate / 100));
END; //
DELIMITER ;
```

Audit Trigger (AFTER UPDATE)

This trigger runs after a successful update to log the changes into the SALARY_LOG table.

```
DELIMITER //
CREATE TRIGGER trg_log_salary_change
AFTER UPDATE ON EMPLOYEE
FOR EACH ROW
BEGIN
    IF OLD.UpdatedSalary <> NEW.UpdatedSalary OR (OLD.UpdatedSalary IS NULL AND NEW.UpdatedSalary IS NOT NULL) THEN
        INSERT INTO SALARY_LOG (EmpID, OldSalary, NewSalary, Note)
        VALUES (NEW.EmpID, OLD.BasicSalary, NEW.UpdatedSalary, CONCAT('Increment of ', NEW.IncrementRate, '% applied'));
    END IF;
END; //
DELIMITER ;
```

Inserting 8 Employee Records

Run this to populate your table with the initial data.

```
INSERT INTO EMPLOYEE (EmpID, EmpName, BasicSalary, StartDate, NoOfPub) VALUES
(101, 'Alice Smith', 50000, '2020-01-15', 5),
(102, 'Bob Johnson', 50000, '2021-05-20', 3),
(103, 'Charlie Brown', 50000, '2022-03-10', 1),
(104, 'David Miller', 50000, '2026-01-01', 10), -- Joined this year (duration < 1)
(105, 'Eve Adams', 60000, '2019-08-12', 0),
(106, 'Frank White', 45000, '2026-02-01', 2),
(107, 'Grace Lee', 70000, '2018-12-25', 6),
(108, 'Henry Ford', 55000, '2020-06-30', 2);
```

Testing and Verification

These updates will fire the triggers. Once finished, run the SELECT queries to see the final output.

```
UPDATE EMPLOYEE SET NoOfPub = 5 WHERE EmpID = 101;
UPDATE EMPLOYEE SET NoOfPub = 3 WHERE EmpID = 102;
UPDATE EMPLOYEE SET NoOfPub = 1 WHERE EmpID = 103;
UPDATE EMPLOYEE SET NoOfPub = 10 WHERE EmpID = 104;


SELECT * FROM EMPLOYEE;
SELECT * FROM SALARY_LOG;
```

4. Results

To verify the triggers, the following select queries were executed:

- SELECT * FROM EMPLOYEE;
- SELECT * FROM SALARY_LOG;

Server: 127.0.0.1

Databases | SQL | Status | User accounts | Export | Import | Settings | Replication | Variables | Charsets | Engines | Plugins

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0185 seconds.)

```
DROP DATABASE IF EXISTS faculty_trigger_lab;
```

[Edit inline] [Edit] [Create PHP code]

Error: #1046 No database selected

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0012 seconds.)

```
CREATE DATABASE faculty_trigger_lab;
```

[Edit inline] [Edit] [Create PHP code]

Error: #1046 No database selected

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
USE faculty_trigger_lab;
```

[Edit inline] [Edit] [Create PHP code]

Error: #1046 No database selected

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0095 seconds.)

```
CREATE TABLE EMPLOYEE ( EmpID INT PRIMARY KEY, EmpName VARCHAR(100) NOT NULL, BasicSalary DECIMAL(10,2) NOT NULL, StartDate DATE NOT NULL, NoOfPub INT NOT NULL CHECK(NoOfPub >= 0), IncrementRate DECIMAL(5,2) DEFAULT 0, UpdatedSalary DECIMAL(10,2) );
```

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0383 seconds.)

```
CREATE TABLE SALARY_LOG ( LogID INT AUTO_INCREMENT PRIMARY KEY, EmpID INT, OldSalary DECIMAL(10,2), NewSalary DECIMAL(10,2), ChangedAt DATETIME DEFAULT CURRENT_TIMESTAMP, Note VARCHAR(100), FOREIGN KEY (EmpID) REFERENCES EMPLOYEE(EmpID) );
```

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0178 seconds.)

```
CREATE TRIGGER trg_calculate_salary BEFORE UPDATE ON EMPLOYEE FOR EACH ROW BEGIN DECLARE duration INT; SET duration = TIMESTAMPDIFF(YEAR, OLD.StartDate, CURDATE()); IF duration > 1 THEN IF NEW.NoOfPub > 4 THEN SET NEW.IncrementRate = 20.00; ELSEIF NEW.NoOfPub IN (2, 3) THEN SET NEW.IncrementRate = 10.00; ELSEIF NEW.NoOfPub = 1 THEN SET NEW.IncrementRate = 5.00; ELSE SET NEW.IncrementRate = 0.00; END IF; ELSE SET NEW.IncrementRate = 0.00; END IF; SET NEW.UpdatedSalary = OLD.BasicSalary + (OLD.BasicSalary * (NEW.IncrementRate / 100)); END;;
```

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0103 seconds.)

```
CREATE TRIGGER trg_log_salary_change AFTER UPDATE ON EMPLOYEE FOR EACH ROW BEGIN IF OLD.UpdatedSalary <> NEW.UpdatedSalary OR (OLD.UpdatedSalary IS NULL AND NEW.UpdatedSalary IS NOT NULL) THEN INSERT INTO SALARY_LOG (EmpID, OldSalary, NewSalary, Note) VALUES (NEW.EmpID, OLD.BasicSalary, NEW.UpdatedSalary, CONCAT('Increment of ', NEW.IncrementRate, '% applied')); END IF; END;;
```

[Edit inline] [Edit] [Create PHP code]

8 rows inserted. (Query took 0.0034 seconds.)

```
INSERT INTO EMPLOYEE (EmpID, EmpName, BasicSalary, StartDate, NoOfPub) VALUES (101, 'Alice Smith', 50000, '2020-01-15', 5), (102, 'Bob Johnson', 50000, '2021-05-20', 3), (103, 'Charlie Brown', 50000, '2022-03-10', 1), (104, 'David Miller', 50000, '2026-01-01', 10), -- Joined this year (duration < 1) (105, 'Eve Adams', 60000, '2019-08-12', 0), (106, 'Frank White', 45000, '2026-02-01', 2), (107, 'Grace Lee', 70000, '2018-12-25', 6), (108, 'Henry Ford', 55000, '2020-06-30', 2);
```

[Edit inline] [Edit] [Create PHP code]

1 row affected. (Query took 0.0033 seconds.)

```
UPDATE EMPLOYEE SET NoOfPub = 5 WHERE EmpID = 101;
```

[Edit inline] [Edit] [Create PHP code]

1 row affected. (Query took 0.0054 seconds.)

```
UPDATE EMPLOYEE SET NoOfPub = 3 WHERE EmpID = 102;
```

[Edit inline] [Edit] [Create PHP code]

1 row affected. (Query took 0.0023 seconds.)

```
UPDATE EMPLOYEE SET NoOfPub = 1 WHERE EmpID = 103;
```

[Edit inline] [Edit] [Create PHP code]

1 row affected. (Query took 0.0024 seconds.)

```
UPDATE EMPLOYEE SET NoOfPub = 10 WHERE EmpID = 104;
```

[Edit inline] [Edit] [Create PHP code]

Showing rows 0 - 7 (8 total, Query took 0.0006 seconds.)

```
SELECT * FROM EMPLOYEE;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 7 (8 total, Query took 0.0006 seconds.)

```
SELECT * FROM EMPLOYEE;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	BasicSalary	StartDate	NoOfPub	IncrementRate	UpdatedSalary
Edit Copy Delete	101	Alice Smith	50000.00	2020-01-15	5	20.00	60000.00
Edit Copy Delete	102	Bob Johnson	50000.00	2021-05-20	3	10.00	55000.00
Edit Copy Delete	103	Charlie Brown	50000.00	2022-03-10	1	5.00	52500.00
Edit Copy Delete	104	David Miller	50000.00	2026-01-01	10	0.00	50000.00
Edit Copy Delete	105	Eve Adams	60000.00	2019-08-12	0	0.00	NULL
Edit Copy Delete	106	Frank White	45000.00	2026-02-01	2	0.00	NULL
Edit Copy Delete	107	Grace Lee	70000.00	2018-12-25	6	0.00	NULL
Edit Copy Delete	108	Henry Ford	55000.00	2020-06-30	2	0.00	NULL

Check all *With selected:* Edit Copy Delete Export

Edit Copy Delete	107	Grace Lee	70000.00	2018-12-25	6	0.0
Edit Copy Delete	108	Henry Ford	55000.00	2020-06-30	2	0.0

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Showing rows 0 - 3 (4 total, Query took 0.0003 seconds.)

```
SELECT * FROM SALARY_LOG;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25

Extra options

```
SELECT * FROM SALARY_LOG;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

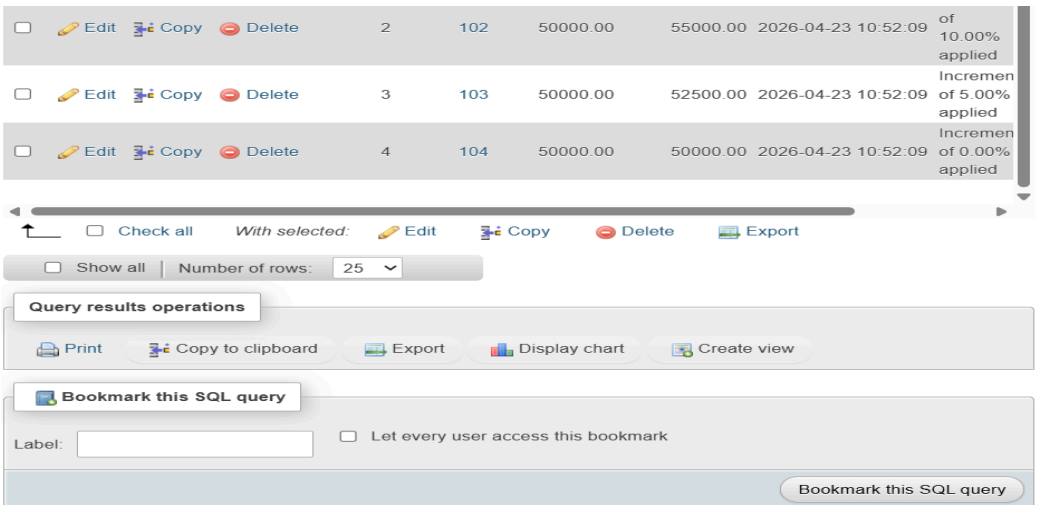
Show all | Number of rows: 25

Extra options

	LogID	EmpID	OldSalary	NewSalary	ChangedAt	Note
Edit Copy Delete	1	101	50000.00	60000.00	2026-04-23 10:52:09	Incremen of 20.00% applied
Edit Copy Delete	2	102	50000.00	55000.00	2026-04-23 10:52:09	Incremen of 10.00% applied
Edit Copy Delete	3	103	50000.00	52500.00	2026-04-23 10:52:09	Incremen of 5.00% applied
Edit Copy Delete	4	104	50000.00	50000.00	2026-04-23 10:52:09	Incremen of 0.00% applied

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25



☐	Edit	Copy	Delete	2	102	50000.00	55000.00	2026-04-23 10:52:09	of 10.00% applied
☐	Edit	Copy	Delete	3	103	50000.00	52500.00	2026-04-23 10:52:09	Incremen of 5.00% applied
☐	Edit	Copy	Delete	4	104	50000.00	50000.00	2026-04-23 10:52:09	Incremen of 0.00% applied

5. Conclusion

The implementation successfully automates salary adjustments based on employee performance and tenure. The use of triggers ensures that the UpdatedSalary is always consistent with the company policy without requiring manual calculations in the application layer.

6. GitHub Repository

https://github.com/shahrimaesha/242002017-CSE210-242D3-LabReport04-Constraints-Alter Untitled-